



EAST CHINA NORMAL UNIVERSITY
SCHOOL OF MATHEMATICAL SCIENCES

Mathematics Syllabus

Spring Algebraic Number Theory Seminar

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Semester: First Semester, 2026–2027

Prerequisites: Abstract Algebra (I), Galois Theory

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Course Description

This seminar offers an introductory exploration of algebraic number theory, a central branch of modern number theory that studies algebraic numbers, number fields, and their arithmetic properties. We will begin with a review of basic commutative algebra, then proceed to the structure of algebraic integers, ideals and ideal classes in Dedekind domains, and the geometry of numbers. The core of the seminar is an in-depth study of the two major themes of classical algebraic number theory: ideal theory and valuation theory, primarily following Cheng Chuangxun's *Algebraic Number Theory*. Participants are expected to present proofs, work through examples, and engage actively in discussions. The goal is to build a solid foundation for further study and research in number theory.

Learning Objectives

- (1) **Master Core Concepts:** Understand the fundamental objects of algebraic number theory—including algebraic integers, Dedekind domains, ideal class groups, unit groups, and valuations with completions—and grasp how they interrelate to form a coherent conceptual framework.
- (2) **Develop Technical Proficiency:** Be able to analyze the arithmetic properties of number fields using both the language of ideals and the language of valuations. Carry out explicit computations such as prime ideal decomposition, class number calculation, and unit group determination in concrete fields (e.g., quadratic and cyclotomic fields).
- (3) **Strengthen Proof Skills:** Through presenting and peer reviewing, be able to independently explain the proofs of central theorems (e.g., unique factorization of ideals, Dirichlet's Unit Theorem, Minkowski's Theorem) and identify their key ideas and subtle points.

Course Content and Schedule

Part I: Algebraic Integers and Dedekind Domains (6 contact hours)

- Number fields and rings of algebraic integers; trace and norm; discriminant; equivalent characterizations of Dedekind domains; unique factorization of ideals; fractional ideals and the ideal group.

Part II: Ideal Decomposition and Ideal Class Groups (6 contact hours)

- Decomposition of prime ideals in extensions: ramification, inertia, and splitting; decomposition patterns in Galois extensions; definition and finiteness of the ideal class group; class number computations for quadratic and cyclotomic fields; the Minkowski bound.

Part III: Unit Groups and Dirichlet's Unit Theorem (4 contact hours)

- Algebraic structure of unit groups; real and complex embeddings; logarithmic embedding and lattices; proof of Dirichlet's Unit Theorem; computation of fundamen-

tal units.

Part IV: Quadratic Forms and the Local-Global Principle (4 contact hours)

- Equivalence and representation of quadratic forms over global fields; brief introduction to Witt cancellation; representability over real and p -adic fields; statement and proof sketch of the Hasse–Minkowski theorem; the philosophy and extensions of the local-global principle.

Part V: Valuation Theory (6 contact hours)

- Absolute values and valuations; Ostrowski’s theorem; completions and p -adic numbers; local and global fields; Hensel’s lemma and its applications.

Part VI: Harmonic Analysis on Local Fields (6 contact hours)

- Topological structure and Haar measure on local fields; additive characters and self-duality; local Fourier transform; local zeta functions and functional equations; connections with L -functions of quadratic forms.

Assessment Methods

This seminar employs process-oriented assessment with no final written examination.

Textbook

J.Neukirch. Algebraic number theory. Springer, 1999.

References

- [1] M.F.Atiyah, I.G.MacDonald. Introduction to Commutative Algebra. Avalon Publishing, 1994.
- [2] Marcus, Daniel A. Number Fields. Springer, 1977.
- [3] Lang, Serge. Algebraic Number Theory. 2nd ed. Springer, 1994. (GTM 110)
- [4] Gerald B. Folland. A Course in Abstract Harmonic Analysis. 2nd ed., CRC Press, 2016.